SoftFuse-KGC: Soft Support and Confidence-Aware Fuzzy Retrieval for Biomedical Knowledge Graph Completion

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Abstract

Biomedical knowledge graph completion can support drug repurposing, but graph-augmented large language model (LLM) pipelines are sensitive to poorly ordered candidates and noisy query-specific evidence. In this paper, we present SoftFuse-KGC, an evidence-calibration framework for drug-focused link prediction with graph-augmented LLMs. The soft-support module reranks top-20 candidates using graded evidence contact, mechanism-oriented support, and penalties for direct shortcuts and contradictory signals. Its confidence-aware fuzzy retriever selects a compact evidence graph while preserving candidate order. We evaluate the framework on PrimeKG, PharmKG, Hetionet, the Drug Repurposing Knowledge Graph (DRKG), and repoDB. On PrimeKG, SoftFuse-KGC raises candidate mean reciprocal rank at 20 (MRR@20) from 0.0646 to 0.1253 while reducing the average graph from 59.93 to 32.34 triples. The corresponding adjusted end-to-end MRR@20 increases from 0.0477 to 0.0747; this ranking measure is reported separately from generated-answer accuracy. Shared-feature learned calibration and leave-one-dataset-out coefficient transfer improve candidate MRR point estimates over the original support formula on four and five datasets, respectively, but do not consistently outperform equal weights or improve exact generation. Matched-budget comparisons show that fuzzy retrieval retains source-covered candidate incidence on all five datasets and more three-hop reachability than uniform random selection on PharmKG, DRKG, and repoDB. Path retention and downstream benefits remain topology dependent. These results characterize complementary roles for candidate calibration and evidence compression. Two PrimeKG case studies and a web interface support inspection of candidate scores, ranking changes, and retained evidence; they do not constitute clinical validation.

Keywords: Biomedical knowledge graph · Knowledge graph completion · Drug repurposing · Soft evidence modeling · Fuzzy retrieval

1 Introduction

Knowledge graphs (KGs) represent entities and their typed relations in a form that supports integrating, analyzing, and reasoning over heterogeneous knowledge [1–3]. This representation is particularly useful in biomedicine, where relevant evidence is distributed across drugs, diseases, genes, proteins, pathways, adverse events, phenotypes, and molecular mechanisms. Biomedical knowledge graphs (BKGs) and related resources, including Hetionet [4], PrimeKG [5], PharmKG [6], DRKG [7], repoDB [8], and others organize this evidence into computable graph structures for translational discovery, drug repurposing,

safety modeling, and therapeutic hypothesis generation. These resources integrate biomedical terminologies and molecular databases such as UMLS [9], DisGeNET [10], UniProt [11], OMIM [12], DrugBank [13], STITCH [14], and STRING [15]. Despite their broad coverage, curated BKGs remain incomplete. Biomedical facts may be missing from databases, dispersed across publications, represented with inconsistent identifiers, or not yet known.

Knowledge graph completion (KGC), often formulated as link prediction, addresses this incompleteness by ranking plausible missing triples from the observed graph [1, 2, 16]. Many established KGC methods rely primarily on graph structure. Translation and semantic matching models such as TransE [17], DistMult [18], ComplEx [19], and RotatE [20] learn representations of entities and relations and use them to score candidate triples efficiently. Graph neural networks such as R-GCN [21] and CompGCN [22], together with relation-aware graph attention models [23], extend this approach by aggregating information from multirelational neighborhoods, allowing local graph context and multi-hop biomedical evidence to influence prediction. Rule-based and path-based methods provide more interpretable predictions through relational patterns and mechanistic paths. However, these methods can become brittle when biomedical evidence is sparse or when the graph does not contain a useful path [24–26].

Textual information provides another source of evidence for biomedical KGC. Entity names, relation labels, and biomedical descriptions contain semantic information that structure-only models may not capture. Language-model-based KGC and text-enhanced ranking methods therefore use relation descriptions, entity names, instruction prompts, retrieval, or candidate-constrained generation to connect graph prediction with biomedical language [27–34]. More recent graph-augmented LLM systems combine candidate retrieval, graph evidence retrieval, graph adapters, and parameter-efficient fine-tuning so that generation can be conditioned more directly on retrieved graph evidence. DrKGC [35] represents this direction by integrating question templates, lightweight candidate retrieval, rule-guided subgraph retrieval, graph-adapter embeddings, and LoRA [36] fine-tuning. For biomedical applications, this design provides a language-based prediction interface while retaining structured evidence from the graph.

The connection between retrieval and generation nevertheless creates two important bottlenecks. First, a candidate-constrained LLM can select only from the entities provided by the candidate retriever. If the gold drug is absent from the top-K candidate list, the generation stage cannot recover it. Even when the gold drug is present, a position near the end of the list may limit its influence on generation and make the prediction less useful for manual review. Second, the graph adapter can use only the evidence contained in the retrieved subgraph. A large or noisy graph may be dominated by direct shortcuts, frequently connected entities, or relations that are structurally close to the query but offer limited clinical or mechanistic information. These limitations are especially relevant to therapeutic head prediction, in which the system ranks drugs or compounds for an incomplete triple of the form $(?, r, \text{disease})$. Evidence for such a query may be indirect and may involve drug targets, disease-associated genes, protein interactions, pathways, previous indications, failed indications, or literature-derived associations.

These retrieval constraints motivate a graded treatment of biomedical evidence. A drug–target edge, disease–gene association, protein interaction, therapeutic association, approved indication, and contraindication-like signal provide different types and strengths of evidence. At the same time, the absence of a path in an incomplete BKG should not automatically be interpreted as negative evidence. Although hard filters are easy to inspect, they may remove recoverable candidates simply because the graph lacks a particular supporting edge. Fuzzy set theory introduced graded membership as an alternative to binary inclusion [37], and fuzzy logic subsequently provided formal tools for reasoning with partial truth and uncertainty [38–41]. Applied to biomedical graph-LLM completion, this perspective suggests that candidate support and graph evidence should be calibrated according to their degree of support, their susceptibility to shortcut behavior, and the confidence associated with the retrieved context.

We address this retrieval–generation interface with SoftFuse-KGC, an evidence-calibration framework for DrKGC-style biomedical graph-augmented LLM completion. SoftFuse-KGC contains two complementary modules. The soft support module reranks retrieved drug candidates using graded evidence contact, mechanism-oriented support, penalties for direct shortcuts, and contradiction cues. The confidence-aware fuzzy retrieval module selects a compact evidence subgraph using candidate rank bands, candidate and query incidence, relation-family priorities, local density, and penalty signals, while preserving the selected candidate order. SoftFuse-KGC is not intended to replace the upstream structural candidate generator. Instead, it evaluates three distinct aspects that are often conflated in graph-LLM pipelines: gold-candidate recoverability, the ordering of recoverable candidates, and the compactness and auditability of the evidence supplied to the graph adapter.

The contribution is the integration and evaluation of explicit biomedical evidence cues at the candidate–graph interface, building on established reranking and subgraph-selection principles. The main contributions are as follows:

- We formulate an inspectable soft-support reranker for a fixed top-20 biomedical candidate set, combining evidence contact with shortcut and contradiction penalties.
- We combine rank-aware edge scoring with coverage-first graph selection and quantify its trade-offs against alternative selectors at matched edge budgets. Preserved candidate incidence is distinguished from preserved paths and representations.
- We evaluate five biomedical datasets, including automatic calibration, coefficient transfer, structural retention, and frozen-model generation, while separating candidate ranking from exact-answer accuracy.
- We provide two PrimeKG case studies across three diagnostic categories and a web application for reproducible inspection of candidate support, rank changes, and selected evidence.

The remainder of this paper is organized as follows. Section 2 reviews structure-based biomedical KGC, graph-augmented LLMs, and soft evidence modeling. Section 3 presents the SoftFuse-KGC framework, including its graph-LLM backbone, soft support module, fuzzy retrieval module, and graph-adapter-enhanced generation. Section 4 describes the study datasets and evaluation metrics before reporting the candidate-stage, end-to-end, case-study, and web-server results. Section 5 discusses the findings and limitations. Section 6 concludes the paper and outlines future research directions.

2 Related Work

2.1 Structure-based biomedical KGC

Translation and semantic matching models such as TransE [17], DistMult [18], ComplEx [19], and RotatE [20] represent entities and relations in continuous embedding spaces and score candidate triples from these representations. Their efficient scoring functions make them useful as large-scale candidate generators, although their predictions are driven mainly by structural regularities in the observed graph. Graph neural networks such as R-GCN [21] and CompGCN [22], together with relation-aware graph attention variants [23], incorporate information from multirelational neighborhoods into entity representations. These models represent local relational context more explicitly than conventional embedding models. Rule-based and path-based approaches instead identify interpretable relational patterns or paths that connect candidate entities to a query. Their dependence on observable paths, however, can make them sensitive to missing evidence [24–26]. Biomedical link prediction and drug repurposing studies further demonstrate that graph proximity must be interpreted carefully because structural closeness alone does not establish therapeutic appropriateness [4, 42–44].

Recent drug-repositioning models also learn how to combine complementary graph views. Zhu et al. [45] combine wave-based spatial propagation with spectral encoding of homogeneous similarity and heterogeneous drug–protein–disease networks, using contrastive alignment to model indirect and long-range associations. FD-MSGL [46] integrates molecular-similarity, interaction, and pathway-regulation graphs

through frequency-domain multi-source learning. These methods operate at the representation-learning and association-prediction stages. They motivate richer evidence features and learned fusion in biomedical KGC, whereas SoftFuse-KGC examines how an already retrieved candidate set and local graph should be organized for a graph-augmented LLM. The present experiments do not compare predictive performance directly against these models.

2.2 Text and graph-augmented LLMs

Text-enhanced and LLM-based KGC methods use relation descriptions, entity names, instruction tuning, retrieval, or candidate-constrained generation to complement structural graph information [27–34]. This approach is relevant to BKGs because biomedical entity names and relation labels contain domain-specific semantic information. Text-only prompting, however, may omit important graph structure, whereas unconstrained generation can produce entities that do not belong to the graph or the intended candidate set.

DrKGC [35] is particularly relevant to the present study because it combines question templates, lightweight candidate retrieval, rule-guided dynamic subgraph retrieval, graph-adapter embeddings, and LoRA [36] fine-tuning. Related biomedical graph-LLM research also shows the value of combining textual representations with domain-specific graph evidence [47]. Nevertheless, a DrKGC-style system remains dependent on two retrieved inputs: the ordered candidate list and the query-specific evidence graph supplied to the graph adapter. Errors in either input can propagate to candidate-constrained generation.

More broadly, retrieval-augmented generation treats external evidence as a non-parametric memory for knowledge-intensive tasks, allowing a generator to use retrieved context in addition to information stored in its parameters [48]. Surveys of KG–LLM integration similarly distinguish KG-enhanced LLMs, LLM-augmented KGs, and systems that combine the two components more closely [49]. In biomedical KGC, these research directions highlight a practical problem at the retrieval–generation interface: prioritizing retrieved candidates and allocating the limited evidence budget to the most relevant local triples.

2.3 Evidence reranking and compact subgraph retrieval

Reranking evidence between retrieval and generation is an established design. G-RAG [50] uses graph neural networks and document-semantic connections to rerank retrieved passages. G-Retriever [51] selects relevant graph context through prize-collecting Steiner-tree optimization and conditions an LLM with graph representations. GRAG [52] retrieves textual subgraphs and combines graph and text views for generation. SubgraphRAG [53] learns parallel triple scores with a lightweight multilayer perceptron and directional structural features, allowing the retrieved graph size to vary with the evidence budget. These studies establish the relevance of relational evidence ranking, learned triple selection, and compact graph context before the present work.

SoftFuse-KGC uses these general principles in a specific biomedical completion setting. The candidate reranker scores drug entities within a fixed retrieved set rather than ranking documents or retrieving new entities. The graph selector combines candidate rank bands, query incidence, shortcut/contradiction cues, and candidate coverage while preserving the selected candidate order. Its "fuzzy" scores are graded support heuristics, not calibrated probabilities of therapeutic validity or a new fuzzy-inference formalism. The empirical question is whether this transparent combination is useful at the graph-adapter interface, how well its coefficients transfer, and which structural information its edge selection retains. Our matched-budget selectors isolate local selection policies; they are not implementations of the complete G-Retriever, GRAG, or SubgraphRAG systems.

2.4 Soft evidence for graph-LLM completion

The present work addresses this interface between a structural candidate source and the graph-LLM input package. Its two central tasks are to promote well-supported candidates within a fixed candidate set and to retain informative evidence triples for downstream graph-enhanced generation.

Soft evidence modeling is appropriate for this setting because biomedical graph relations are heterogeneous. A drug–target edge, disease–gene association, protein interaction, literature-derived drug–disease association, and approved indication do not provide equivalent therapeutic evidence. Hard filtering remains interpretable, but it can be unreliable in an incomplete BKG: The absence of an observed supporting relation does not necessarily indicate the absence of biomedical support. Fuzzy set theory represents membership as a degree rather than a binary decision [37], while fuzzy logic provides foundations for reasoning with partial truth and uncertainty [38–41]. Fuzzy knowledge graphs have also been combined with complex fuzzy inference systems to support approximate reasoning when the available knowledge or rule base is limited [54]. In clinical decision support, picture fuzzy sets and knowledge-based reasoning have been used to represent imprecise diagnostic information [55, 56]. SoftFuse-KGC applies the same general principle to evidence calibration in biomedical graph-LLM completion. Observed relations, missing support, indirect paths, shortcut signals, and contradiction-like evidence remain distinguishable instead of being collapsed prematurely into a single binary keep-or-discard decision.

3 Methods

3.1 DrKGC-style graph-LLM backbone

The backbone retains the central components of DrKGC [35]: candidate retrieval, source-subgraph retrieval, graph-adapter embedding enhancement, and candidate-constrained LLM prompting. Across all datasets, we use a lightweight R-GCN [21] ranker as the graph-compatible source. This choice is motivated by interface compatibility rather than by an assumption that R-GCN is the strongest standalone link predictor. R-GCN provides relation-aware entity embeddings and local neighborhoods in the format expected by the DrKGC-style subgraph encoder. Using the same source also keeps the graph-adapter interface fixed and makes the calibration modules comparable across datasets.

The calibration modules themselves are not tied to R-GCN. Any structural or hybrid retriever can serve as the source if it provides a fixed candidate list, compatible entity identifiers or embeddings, and a query-specific subgraph. Integrating stronger compatible sources, such as ComplEx [19], DistMult [18], relation-aware attention retrievers [23], or hybrid retrievers, remains future work. The initial candidate list is defined as follows:

$$C_q^{(0)} = (c_1, c_2, \ldots, c_K), \quad K = 20, \tag{1}$$

where the backbone structural score determines the order. Candidate names are inserted explicitly into the prompt, which instructs the LLM to return one entity from this set. The source subgraph is intentionally larger than the final fuzzy graph. This design preserves the original DrKGC graph-adapter interface while retaining enough local evidence for SoftFuse-KGC to determine which candidate and edge signals should be retained or downweighted.

3.2 Overview of the SoftFuse-KGC framework

SoftFuse-KGC operates as an evidence-calibration layer between candidate retrieval and graph-adapter-enhanced generation. Fig. 1 shows the four implementation stages. **A. Biomedical backbone preparation** constructs the task-specific BKG, trains or loads the graph-compatible drug ranker, exports global entity embeddings, and prepares path rules and question templates. **B. Candidate and evidence retrieval** receives a therapeutic query of the form $(?, r_q, \text{disease})$, retrieves the top-20 drug candidates, and packages a source subgraph containing direct candidate–query edges, candidate neighborhoods, and rule-compatible local paths. **C. SoftFuse-KGC evidence calibration** applies the two proposed modules: C1 soft support reranks candidates from graded support and penalty cues, whereas C2 confidence-aware fuzzy retrieval selects a smaller evidence graph using rank-band confidence and edge-level evidence quality. **D. Graph-enhanced candidate-constrained generation** passes the calibrated candidates and selected graph to the graph adapter and LoRA-tuned LLM, which must return one supplied candidate.

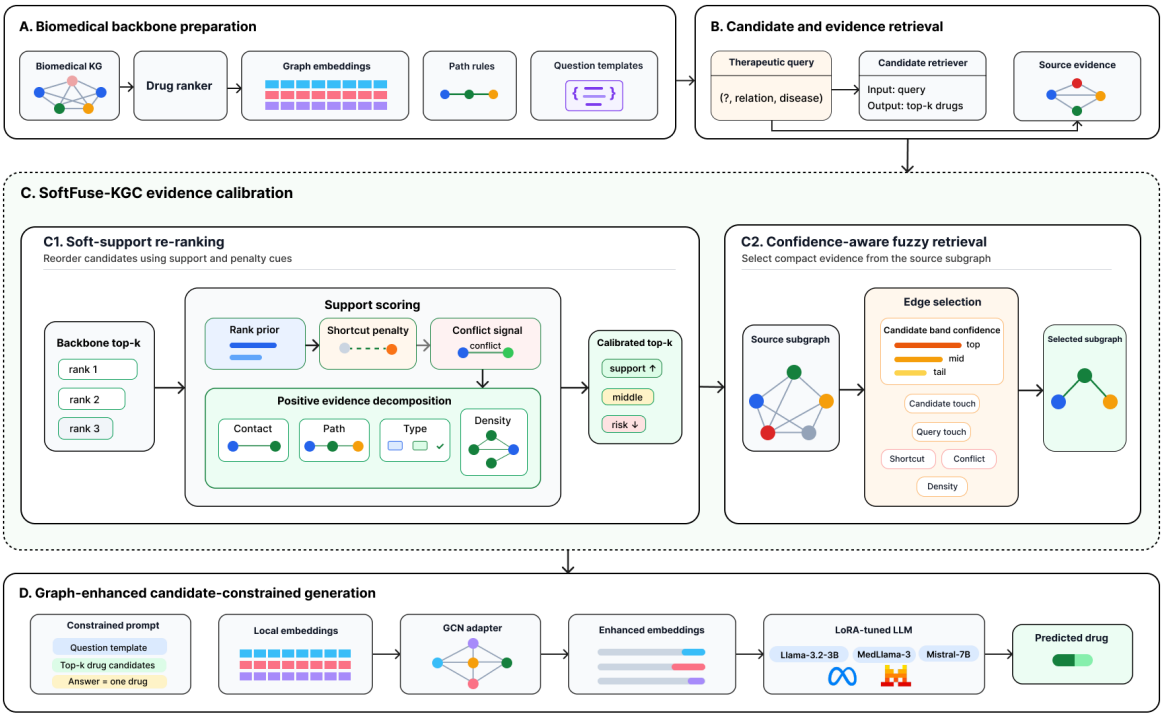


Fig. 1 SoftFuse-KGC framework for evidence-calibrated biomedical graph-augmented LLM completion. A. Biomedical backbone preparation builds the BKG, graph-compatible drug ranker, graph embeddings, path rules, and question templates. **B. Candidate and evidence retrieval** converts a therapeutic query into a top-20 candidate list and source evidence subgraph. **C. SoftFuse-KGC evidence calibration** includes C1 soft support reranking, which combines a rank prior, decomposed positive support, shortcut penalties, and contradiction signals, and C2 confidence-aware fuzzy retrieval, which selects compact evidence using candidate bands, candidate and query incidence, shortcut, conflict, and density cues. **D. Graph-enhanced candidate-constrained generation** passes the calibrated candidates and selected graph through the graph adapter and LoRA-tuned LLM to predict one drug

The distinction between C1 and C2 is central to the study design. We attribute changes in candidate order to soft support and changes in graph size or evidence content to fuzzy retrieval.

3.3 Soft support candidate reranking

The soft support module reranks the backbone candidates using evidence extracted from the initial source subgraph $S_q^{(0)}$. This is the graph package constructed by the DrKGC-style retriever before fuzzy compression. It contains candidate–query paths, candidate-neighborhood edges, and relation-specific evidence that may provide support, indicate risk, or create an ambiguous connection between a candidate and the query. Soft support converts these heterogeneous signals into a calibrated candidate score rather than making a binary keep-or-discard decision.

We formulate soft support as a unified candidate score. It combines positive evidence $E_i(q)$, direct-shortcut or popularity evidence $D_i(q)$, contradiction-like evidence $X_i(q)$, and, where selected, a backbone rank prior. Dataset-specific instantiations set the coefficients of unavailable or inactive terms to zero.

For query q and candidate c_i, the positive support family is computed as follows:

$$\begin{aligned} E_i(q) =& \omega_{\text{touch}} T_i(q) + \omega_{\text{path}} P_i(q) \\ &+ \omega_{\text{schema}} A_i(q) + \omega_{\text{density}} \Delta_i(q), \end{aligned} \tag{2}$$

where $T_i(q)$ denotes candidate evidence contact, $P_i(q)$ denotes mechanism-oriented path support, $A_i(q)$ denotes type or schema compatibility, and $\Delta_i(q)$ denotes local evidence density. A contact-based instantiation uses $T_i(q) = \mathbb{I}[\deg_{S_q^{(0)}}(c_i) > 0]$, where $\deg_{S_q^{(0)}}(c_i)$ counts source-subgraph triples incident to the candidate. In this setting, $P_i(q)$ summarizes shared-gene overlap, gene–gene bridge evidence, and contacts

with candidate- and query-side mechanism neighborhoods:

$$\begin{aligned} P_i(q) =& \nu_s M_i^s(q) + \nu_b M_i^b(q) \\ &+ \nu_c T_i^c(q) + \nu_q T_i^q(q), \\ M_i^s(q) =& \min\left(1, \frac{|G(c_i) \cap G(q)|}{\eta_s}\right), \\ M_i^b(q) =& \min\left(1, \frac{B_i(q)}{\eta_b}\right), \end{aligned} \tag{3}$$

where $G(c_i)$ is the set of target or associated genes for the compound, $G(q)$ is the disease-gene set, and $B_i(q)$ counts gene–gene bridge edges between these sets. The terms $T_i^c(q)$ and $T_i^q(q)$ indicate whether the source subgraph contains candidate-side and query-side mechanism neighborhoods, respectively. Schema support is defined as follows:

$$A_i(q) = \mathbb{I}[\tau(c_i) \in \mathcal{T}_{\text{drug}}]\, \mathbb{I}[\tau(t_q) \in \mathcal{T}_{\text{disease}}], \tag{4}$$

when entity-type annotations are available. The density term is a bounded local count, for example,

$$\Delta_i(q) = \min\left(1, \frac{\log(1 + \deg_{S_q^{(0)}}(c_i) + \deg_{S_q^{(0)}}(t_q))}{\eta_\Delta}\right). \tag{5}$$

Together, these components make $E_i(q)$ a positive evidence score rather than a detector of a single direct link.

The two penalty terms represent different failure modes. The first, $D_i(q)$, captures a direct shortcut or candidate-level popularity:

$$D_i(q) = \max\left(\mathbb{I}[c_i \leftrightarrow t_q \text{ in } S_q^{(0)}], \min\left(1, \frac{\deg_{\text{direct}}(c_i)}{\eta_D}\right)\right), \tag{6}$$

The first component detects a direct candidate–query edge in the source graph. The second penalizes candidates that act as generic hubs across direct disease relations. The contradiction term $X_i(q)$ marks contraindication, conflict, failed, or suspended-indication evidence:

$$X_i(q) = \mathbb{I}[c_i \leftrightarrow t_q \text{ through contradiction-like evidence}]. \tag{7}$$

We first compute an evidence-only score and then, when selected, combine its within-query normalized value with the backbone rank prior:

$$\begin{aligned} \widetilde{S}_{\theta_d}(c_i, q) =& \alpha_{E,d} E_i(q) - \alpha_{D,d} D_i(q) - \alpha_{X,d} X_i(q), \\ S_{\theta_d}(c_i, q) =& \mu_d R_i(q) + (1 - \mu_d) N_q\Big(\widetilde{S}_{\theta_d}(c_i, q)\Big). \end{aligned} \tag{8}$$

Here, $R_i(q) = (K - r_i^{(0)} + 1)/K$ is the normalized backbone rank prior. The function $N_q(\cdot)$ is the within-query normalization when a rank-prior mixture is selected and the identity map otherwise. The index d denotes the dataset, and θ_d collects its calibration parameters.

Candidates are sorted by decreasing soft support score, with the original backbone rank used as a stable tiebreaker:

$$c_i \prec c_j \iff S_i > S_j \quad \text{or} \quad (S_i = S_j \text{ and } r_i^{(0)} < r_j^{(0)}), \tag{9}$$

where $S_i = S_{\theta_d}(c_i, q)$ and $r_i^{(0)} = \text{rank}_0(c_i)$. We denote the resulting candidate list by C_q^{soft}.

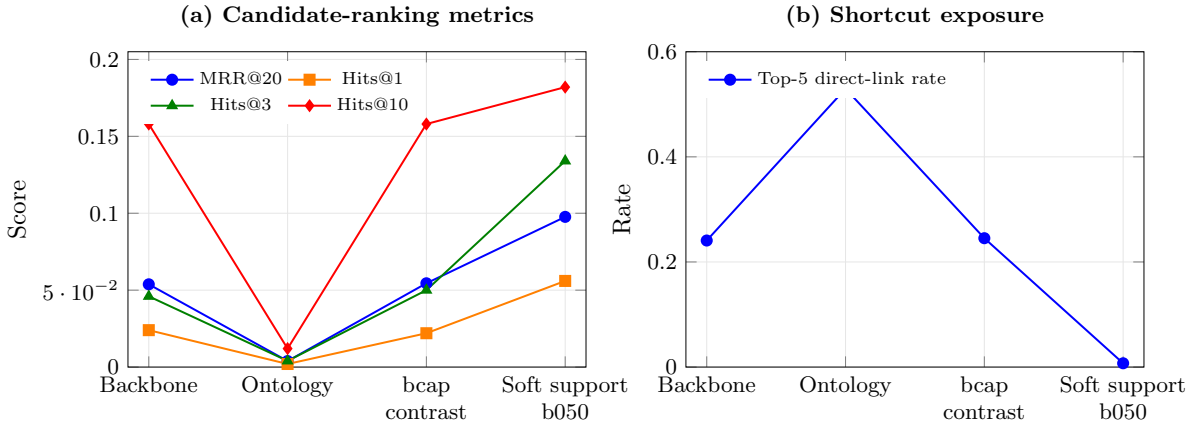


Fig. 2 Soft-support calibration and shortcut exposure on PrimeKG validation. Panel (a) compares the raw source, hard ontology filtering, capped-evidence contrast, and selected b050 setting. MRR@20 assigns zero reciprocal rank to absent gold candidates. Panel (b) reports top-five direct-link exposure. The selected contact-based score raises validation MRR from 0.0538 to 0.0976 while reducing direct-link exposure from 0.2408 to 0.0072.

Online Resource 1 provides the feature-selection procedure, weight-assignment method, saturation constants, normalization choices, and validation-frozen coefficients for equations (2)–(8). Supplementary Section S1 describes the soft-support feature audit and coefficient assignment, Table S1 reports the dataset-specific instantiations, and Supplementary Section S6 summarizes the PrimeKG formula variants used as representative ablations.

Fig. 2 presents a validation-level view of this behavior on PrimeKG. Panel A compares candidate-ranking metrics for the raw backbone, ontology filtering, a contrastive support variant, and the selected soft support setting. Ontology filtering serves as the hard-filter control: candidates are retained only when they satisfy binary type, schema, or ontology-support conditions, while candidates without such support are discarded or demoted. This control is diagnostically useful but can be brittle in an incomplete BKG because missing ontology evidence is not equivalent to negative biomedical evidence. By contrast, the selected soft support setting retains all top-20 candidates and changes only their order. Panel B shows that this setting improves validation ranking while lowering the top-five direct-link rate. The comparison therefore supports the interpretation that the gain reflects graded evidence calibration rather than the promotion of shortcut-connected candidates.

3.4 Confidence-aware fuzzy retrieval

The fuzzy retrieval module changes the evidence graph supplied to the graph adapter while preserving a validation-selected candidate order. Let C_q^{in} denote this fixed input list. In the main pipeline, it is usually the soft-support list; the protocol can instead use a backbone-compatible list when validation analysis indicates that soft support should remain diagnostic rather than define the main row. Let $S_q^{(0)}$ denote the source evidence subgraph for query q, and let $e = (u, r, v)$ be an evidence triple. Fuzzy retrieval assigns a confidence-aware score to every edge and selects a compact subgraph S_q^{fuzzy} under a dataset-specific graph budget.

A key component is candidate rank-band confidence. Under the fixed top-20 protocol, the selected list is partitioned into ordered bands for the highest-ranked, middle, and recoverable tail candidates. When an edge touches one or more candidates, its band score is the largest weight assigned to any touched candidate:

$$B(e, q) = \max_{c_i \in C_q^{\text{in}} : e \sim c_i} w_b(r_i), \tag{10}$$

and we set $B(e, q) = 0$ when e touches no candidate. Here, $e \sim c_i$ means that edge e is incident to candidate c_i or is annotated as part of a candidate-anchored evidence path; r_i is the selected rank of c_i.

The confidence-aware edge score combines rank-band confidence $B(e,q)$, candidate incidence $T_e^c(q)$, query incidence $T_e^q(q)$, a top-candidate bonus $H_e(q)$, a direct-shortcut penalty $D_e(q)$, a contradiction or failed-evidence penalty $X_e(q)$, local density $\rho_e(q)$, and relation-family priority $F_r(e)$. Dataset-specific instantiations give zero coefficients to unavailable or inactive terms.

The full edge score is computed as follows:

$$\begin{aligned} S_{\text{edge}}(e,q) =& \lambda_f F_r(e) + \lambda_b B(e,q) + \lambda_c T_e^c(q) \\ &+ \lambda_q T_e^q(q) + \lambda_{\text{top}} H_e(q) - \lambda_d D_e(q) \\ &- \lambda_x X_e(q) + \lambda_\rho \rho_e(q), \end{aligned} \tag{11}$$

where

$$\begin{aligned} T_e^c(q) &= \mathbb{I}[u \in C_q^{\text{in}} \text{ or } v \in C_q^{\text{in}}], \\ T_e^q(q) &= \mathbb{I}[u = t_q \text{ or } v = t_q], \\ H_e(q) &= \mathbb{I}[\exists c_i \in C_q^{\text{in}} : \ e \sim c_i \text{ and } r_i \le K_{\text{top}}], \end{aligned} \tag{12}$$

For edge $e = (u, r, v)$, K_{top} denotes the upper boundary of the top-candidate band. The term $F_r(e)$ encodes relation-family priority, $D_e(q)$ captures direct-shortcut evidence, and $X_e(q)$ captures contraindication, conflict, failed, or suspended-indication evidence. The density term is computed as follows:

$$\rho_e(q) = \frac{\log(1 + \deg_{S_q^{(0)}}(u)) + \log(1 + \deg_{S_q^{(0)}}(v))}{\eta_\rho}, \tag{13}$$

and is clipped to a bounded range in the implementation.

The selected graph is represented as follows:

$$S_q^{\text{fuzzy}} = \text{TopEdges}\left(S_q^{(0)}, S_{\text{edge}}, \tau_{d,q}\right), \tag{14}$$

where $\tau_{d,q}$ is the dataset- and query-specific graph budget.

Edge selection follows the implementation order of a coverage-then-fill procedure. First, a candidate-coverage pass retains a high-scoring available edge for candidates that would otherwise lose all candidate-touching evidence. At the reported native budgets the output fits the target size; a coverage-first reservation can exceed a smaller requested budget, which is examined separately in Supplementary Section S13. When the prepared graph contains informative query-touching or mechanistic relation families, optional query-coverage and relation-family reservation passes are applied next. The remaining slots are filled in descending order of $S_{\text{edge}}(e,q)$ until the target size is reached. Fuzzy retrieval is therefore designed to produce a smaller, cleaner graph for the adapter and for human inspection, rather than to improve candidate MRR directly.

The fuzzy module preserves the order of its selected candidate list:

$$C_q^{\text{fuzzy}} = C_q^{\text{in}}. \tag{15}$$

Accordingly, candidate-stage changes are attributed to the input ranking, whereas graph-size reduction and evidence selection are attributed to fuzzy retrieval. Candidate coverage here refers to incidence relative to the source subgraph; it does not guarantee preservation of query–candidate paths or biological mechanisms.

Online Resource 1 provides the feature-selection procedure, edge-weight assignment, rank-band definitions, relation-family priorities, top-candidate bonus settings, density normalizers, graph budgets, and coverage checks for equations (10)–(15). Supplementary Section S2 describes the feature audit and edge-weight assignment, Table S2 reports the frozen edge-scoring weights, and Supplementary Section S3 and Table S3 report graph budgets and coverage checks.

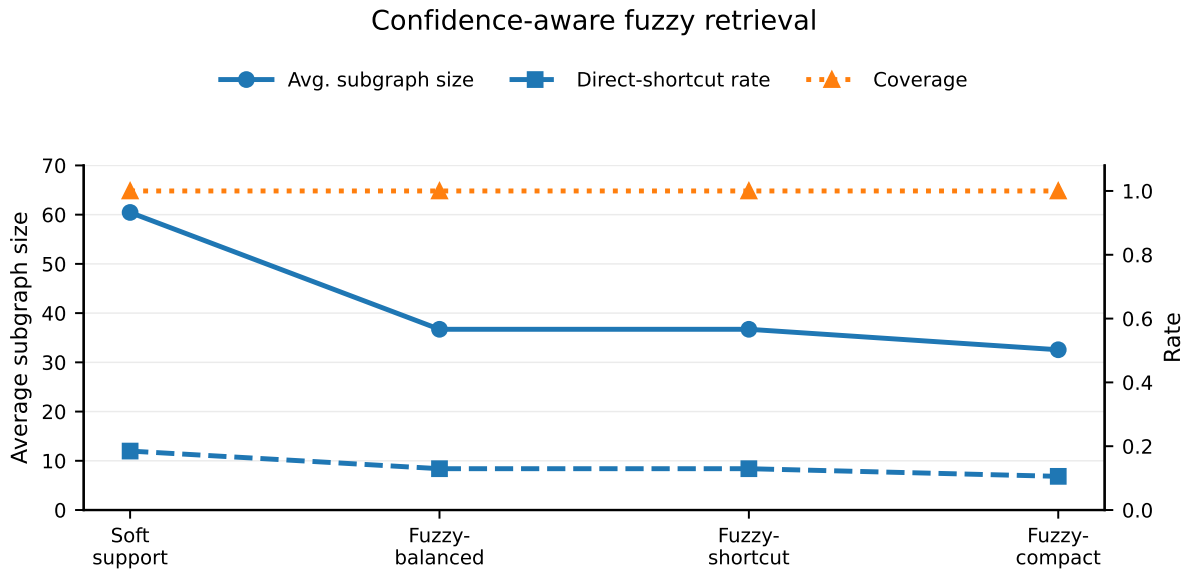


Fig. 3 Confidence-aware fuzzy retrieval selects the compact evidence package used after soft support. The diagnostic plot compares the soft-support source graph with fuzzy retrieval variants in terms of average subgraph size, direct-shortcut exposure, and candidate coverage. Because fuzzy retrieval preserves the soft-support candidate order, the selected variant is chosen for graph efficiency and evidence cleanliness while maintaining Gold@20, MRR@20, and candidate coverage

Fig. 3 summarizes the evidence-selection trade-off. Starting from the selected input package, the fuzzy variants show how much evidence can be removed while the candidate list remains fixed. The selected setting is therefore not chosen for a new candidate-ranking gain. Instead, it preserves Gold@20, candidate MRR, and candidate coverage while supplying a smaller graph to the adapter.

3.5 Graph-adapter-enhanced generation

For each query, the prompt represents the query entity and all candidate entities with special placeholders. The graph enhancer replaces their placeholder embeddings with graph-aware embeddings. Each entity begins with a frozen structural embedding from the lightweight ranker. When a subgraph is available and sufficiently large, a relation-aware graph convolution updates node representations within the query-specific graph. The adapter then projects the concatenation of the global and locally updated embeddings to the hidden dimension of the LLM:

$$\mathbf{z}_e = \text{Adapter}\left([\mathbf{h}_e^{\text{global}}; \mathbf{h}_e^{\text{local}}]\right). \tag{16}$$

If an entity is absent from the retrieved subgraph, its local embedding is set to its global embedding. This fallback prevents missing graph evidence from producing an invalid representation and is consistent with the soft-evidence interpretation.

We fine-tune the LLM through LoRA [36] adapters while training the graph adapter and keeping the structural embeddings frozen. The prompt requires one candidate entity as the answer. Llama 3.2 3B is the primary language model for end-to-end evaluation.

4 Experimental Results

4.1 Datasets and task setting

We represent each BKG as a directed, multirelational graph of biomedical entities and typed relations. Across all five datasets, drug-focused therapeutic completion is formulated as head prediction. Given a query tail t_q, the system ranks a drug or compound head for the incomplete triple $(?, r_q, t_q)$, where r_q is a

Table 1 Summary of the five biomedical evaluation datasets. Train, validation, and test values count target-relation triples. Candidates are the size of the restricted drug or compound universe. Graph edges, entities, and relations describe the prepared training/evidence graph after validation and test target edges were excluded.

Dataset	Target relation	Train	Valid	Test	Candidates	Graph edges	Entities	Relations
PrimeKG	indication	8,388	500	500	1,801	136,351	10,453	4
PharmKG	T proxy	28,960	500	500	1,342	386,768	7,247	28
Hetionet	CtD	555	100	100	1,552	2,249,997	47,031	24
DRKG	DrugBank treats	3,968	500	500	1,542	187,962	13,918	85
repoDB	approved indication	5,677	500	500	1,519	90,804	8,818	63

dataset-specific therapeutic or indication-style relation. Table 1 summarizes the target tasks, data splits, restricted candidate universes, and prepared training/evidence graphs. The graph statistics describe the evidence supplied to the graph-compatible pipeline after target triples assigned to validation or test were excluded; they are not the sizes of the unprocessed source KGs.

PrimeKG [5] is the primary evaluation setting. We derived a drug-restricted indication benchmark and prepared an evidence graph with four relation families: indication, target, associated_with, and protein–protein interaction. This combination supports both direct therapeutic context and mechanism-oriented drug–gene–disease paths. PharmKG [6] provides a transfer setting with a larger relation vocabulary. Its relation T is treated as a therapeutic-association proxy rather than a confirmed clinical-indication label, so results on this dataset indicate transfer across graph semantics rather than direct replication of the PrimeKG task.

Hetionet [4] uses the compound-treats-disease relation CtD, with the candidate universe restricted to compounds. Its evidence graph is much larger than the other prepared graphs and has a distinct topology and relation vocabulary, providing a test of robustness to graph scale and structure. For DRKG [7], we use the DrugBank [13] relation `DRUGBANK::treats::Compound:Disease` as the target. The controlled evidence graph combines selected compound–gene target relations, disease–gene relations, gene–gene relations, and auxiliary compound–disease relations rather than using every raw DRKG edge.

In repoDB [8], approved drug–indication pairs define positive target triples, whereas failed pairs are retained as diagnostic evidence. repoDB disease identifiers are UMLS [9] CUIs that do not map directly to the reused DRKG disease namespace, so they remain local UMLS nodes. The resulting graph combines approved target edges, failed-like diagnostic edges, and DRKG-derived compound–gene and gene–gene evidence. This identifier mismatch produces sparser query-specific mechanistic paths and makes repoDB a useful test of evidence calibration under limited cross-resource alignment.

Together, the five settings vary in target semantics, candidate-universe size, graph density, relation diversity, and availability of mechanism-oriented support. These differences allow the evaluation to distinguish improvements that transfer across datasets from effects that depend on a particular graph topology or evidence family.

4.2 Evaluation metrics

All experiments use the same fixed top-20 protocol. Candidate lists for validation and test queries are generated before scoring, and a gold entity absent from a list receives the sentinel rank 21. We use $K = 20$ because this is the candidate-list size accepted by the graph-LLM backbone. It is also small enough for candidate-constrained generation while supporting informative Hits@1/3/10 and Gold@20 analyses.

Here, Q denotes the set of evaluation queries; y_q is the gold entity for query q; C_q is its fixed candidate list; and $\mathrm{rank}_C(y_q)$ is the one-based position of the gold entity in that list. A present gold entity receives its top-20 position, whereas an absent gold receives rank 21. Reciprocal rank is caculated as follows:

$$\mathrm{RR@20}(q) = \begin{cases} 1/\mathrm{rank}_C(y_q), & \mathrm{rank}_C(y_q) \le 20, \\ 0, & \mathrm{rank}_C(y_q) = 21. \end{cases} \tag{17}$$

The candidate MRR is caculated as follows:

$$\mathrm{MRR@20} = \frac{1}{|Q|} \sum_{q \in Q} \mathrm{RR@20}(q). \tag{18}$$

Hits@K and Gold@20 are caculated as follows:

$$\mathrm{Hits@}K = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{I}\left[\mathrm{rank}_C(y_q) \le K\right], \quad K \in \{1, 3, 10\}, \tag{19}$$

$$\mathrm{Gold@20} = \frac{1}{|Q|} \sum_{q \in Q} \mathbb{I}[y_q \in C_q]. \tag{20}$$

Hits@1, Hits@3, and Hits@10 measure first-choice accuracy, early-rank utility, and broader candidate-list usefulness, respectively. We report Gold@20 separately because candidate coverage is a prerequisite for candidate-constrained generation.

For end-to-end evaluation, the LLM output is normalized and matched against the candidate names. Let $\hat{y}_q$ denote the normalized generated answer. An exact match with the gold answer gives E2E rank 1, whereas an absent gold entity gives E2E rank 21. When the generated answer is incorrect but occurs above the gold entity in the candidate list, the gold rank is unchanged. In all other cases, the gold rank is displaced by one position:

$$r_{\mathrm{e2e}}(q) = \begin{cases} 1, & \hat{y}_q = y_q, \\ 21, & y_q \notin C_q, \\ \mathrm{rank}_C(y_q), & \hat{y}_q \in C_q, \\ & \mathrm{rank}_C(\hat{y}_q) < \mathrm{rank}_C(y_q), \\ \min(21, \mathrm{rank}_C(y_q) + 1), & \text{otherwise.} \end{cases} \tag{21}$$

The end-to-end reciprocal rank is described as below:

$$\mathrm{RR@20}_{\mathrm{e2e}}(q) = \begin{cases} 1/r_{\mathrm{e2e}}(q), & r_{\mathrm{e2e}}(q) \le 20, \\ 0, & r_{\mathrm{e2e}}(q) = 21. \end{cases} \tag{22}$$

Adjusted E2E MRR measures the gold position after accounting for the generated answer and the supplied candidate order. It can remain positive for an incorrect or invalid answer and therefore is not generated-answer accuracy. We additionally report exact normalized answer accuracy, invalid-output rate, and prediction agreement in Supplementary Sections S12–S13. All current module, calibration, and selector comparisons use the same deterministic generation and exact matching protocol within each dataset.

4.3 Candidate stage comparison with structure baselines

The first analysis evaluates candidate retrieval before LLM generation. This stage determines whether candidate-constrained generation can recover the gold drug. A gold entity outside the fixed top-20 list cannot be selected, while a gold entity near the tail remains difficult for both the model and a human reviewer to identify. Table 2 therefore reports three complementary quantities. Gold@20 measures recoverability, MRR@20 measures gold position under the absent-gold rank-21 rule, and Hits@10 indicates whether the gold appears in the upper half of the list.

The table contains two types of rows. The standalone structure-only baselines provide reference candidate-retrieval performance for conventional embedding and graph models. The DrKGC-style source and SoftFuse-KGC rows are graph-compatible: in addition to ranking candidates, they must provide identifiers, embeddings, and query-specific source subgraphs for the graph adapter. Both graph-compatible

Table 2 Fixed top-20 candidate-ranking comparison across five biomedical datasets. All values are test metrics under the same absent-gold rank-21 rule. Metrics marked with (↑) are better when larger. Bold marks the best value for each dataset and metric. SoftFuse-KGC is the main graph-compatible row and should be interpreted as an evidence-calibrated DrKGC-style source, not as a universal replacement for the strongest structure-only candidate generator.

Model	Metric	PrimeKG	PharmKG	Hetionet	DRKG	repoDB
TransE	Gold@20 (↑)	0.104	0.060	0.270	0.270	0.248
	MRR@20 (↑)	0.022	0.009	0.046	0.055	0.053
	Hits@10 (↑)	0.072	0.028	0.150	0.160	0.130
DistMult	Gold@20 (↑)	0.434	**0.102**	0.390	**0.432**	0.482
	MRR@20 (↑)	0.089	0.017	0.072	**0.100**	**0.104**
	Hits@10 (↑)	0.264	0.038	**0.290**	**0.276**	**0.320**
ComplEx	Gold@20 (↑)	**0.574**	0.066	**0.440**	0.406	**0.502**
	MRR@20 (↑)	0.125	0.009	0.090	0.086	0.084
	Hits@10 (↑)	**0.384**	0.036	0.250	0.260	0.312
RotatE	Gold@20 (↑)	0.134	0.024	0.200	0.182	0.202
	MRR@20 (↑)	0.033	0.002	0.043	0.024	0.022
	Hits@10 (↑)	0.074	0.004	0.120	0.098	0.100
R-GCN	Gold@20 (↑)	0.220	0.092	**0.440**	0.200	0.214
	MRR@20 (↑)	0.060	0.020	0.099	0.043	0.048
	Hits@10 (↑)	0.186	0.046	0.220	0.120	0.128
HRGAT	Gold@20 (↑)	0.212	0.094	0.410	0.204	0.222
	MRR@20 (↑)	0.033	0.019	**0.101**	0.042	0.052
	Hits@10 (↑)	0.128	0.044	**0.290**	0.134	0.146
DrKGC-style source	Gold@20 (↑)	0.240	0.092	**0.440**	0.200	0.214
	MRR@20 (↑)	0.065	0.020	0.099	0.043	0.048
	Hits@10 (↑)	0.192	0.046	0.220	0.120	0.128
SoftFuse-KGC	Gold@20 (↑)	0.240	0.092	**0.440**	0.200	0.214
	MRR@20 (↑)	**0.125**	**0.028**	0.095	0.047	0.048
	Hits@10 (↑)	0.222	**0.068**	0.280	0.142	0.128

Table 3 End-to-end module comparison for the graph-LLM pipeline. Gold@20, candidate MRR@20, and E2E MRR@20 are higher-is-better metrics (↑), whereas average graph size is lower-is-better (↓). E2E MRR@20 uses Llama 3.2 3B candidate-constrained generation with the adjusted-rank rule defined in Section 4.2. Bold marks the best row for each dataset and metric where the comparison is meaningful.

Row	Metric	PrimeKG	PharmKG	Hetionet	DRKG	repoDB
DrKGC-style source	Gold@20 (↑)	0.240	0.092	0.440	0.200	0.214
	Candidate MRR@20 (↑)	0.0646	0.0205	**0.0994**	0.0432	**0.0481**
	E2E MRR@20 (↑)	0.0477	0.0156	0.0819	0.0410	0.0366
	Avg. graph (↓)	59.93	100.00	80.00	100.00	100.00
Soft support	Gold@20 (↑)	0.240	0.092	0.440	0.200	0.214
	Candidate MRR@20 (↑)	**0.1253**	**0.0282**	0.0954	**0.0466**	0.0481
	E2E MRR@20 (↑)	**0.0747**	0.0207	**0.1001**	**0.0463**	0.0366
	Avg. graph (↓)	59.93	100.00	80.00	100.00	100.00
Fuzzy retrieval	Gold@20 (↑)	0.240	0.092	0.440	0.200	0.214
	Candidate MRR@20 (↑)	**0.1253**	**0.0282**	0.0954	**0.0466**	**0.0481**
	E2E MRR@20 (↑)	**0.0747**	**0.0210**	**0.1001**	**0.0463**	**0.0378**
	Avg. graph (↓)	**32.34**	**55.00**	**44.00**	**55.00**	**55.00**

rows use the same lightweight R-GCN-compatible source. Their difference therefore reflects evidence calibration rather than replacement of the upstream retriever.

PrimeKG provides the clearest candidate-stage result. SoftFuse-KGC increases MRR@20 from 0.065 to 0.125 while Gold@20 remains 0.240, showing that soft support reorders recoverable candidates rather than expanding the set. PharmKG follows the same direction in the transfer setting: MRR@20 rises from 0.020 to 0.028 and Hits@10 from 0.046 to 0.068, with Gold@20 unchanged at 0.092. DRKG shows a smaller positive MRR@20 change, from 0.043 to 0.047. Hetionet is mixed. Gold@20 remains 0.440 and Hits@10 increases from 0.220 to 0.280, but MRR@20 decreases from 0.099 to 0.095. Soft support therefore improves some upper-half placements while moving a small number of first-rank cases downward. repoDB is neutral at the candidate stage, consistent with its local UMLS disease nodes and sparse query-specific mechanistic support.

The standalone results provide context for these changes. ComplEx achieves the highest PrimeKG Gold@20; HRGAT and R-GCN are strong on Hetionet; and DistMult has the highest candidate MRR@20 on DRKG and repoDB. SoftFuse-KGC should therefore not be interpreted as a replacement for the strongest structure-only retriever. Its specific role is to calibrate the graph-compatible candidate package passed to the graph-LLM pipeline. The next analysis tests whether candidate calibration and evidence selection are reflected in end-to-end behavior.

4.4 End-to-end graph-LLM pipeline comparison

Table 3 compares the three graph-compatible pipeline rows after candidate-constrained generation: the raw DrKGC-style source, the soft-support row, and the confidence-aware fuzzy retrieval row. This system-level comparison reflects both inputs available to the downstream model: an ordered candidate list and a query-specific graph package delivered through the adapter. It also isolates the proposed modules. Soft support changes candidate order, whereas fuzzy retrieval preserves that order and changes the evidence graph.

PrimeKG shows a clear candidate-ranking improvement accompanied by higher adjusted E2E MRR@20: soft support changes the latter from 0.0477 to 0.0747. Fuzzy retrieval retains 0.0747 to four decimal places while reducing the average input from 59.93 to 32.34 edge occurrences. This count reduction includes duplicate removal, as quantified in Section 4.6. On PharmKG, adjusted E2E MRR changes from 0.0156 for the raw source to 0.0207 with soft support and 0.0210 with fuzzy retrieval, while the graph shrinks from 100 to 55 triples. These are changes in the adjusted ranking metric, not equivalent improvements in exact generation.

On Hetionet, soft support increases adjusted E2E MRR@20 from 0.0819 to 0.1001 despite a slightly lower candidate MRR than the raw source. Fuzzy retrieval retains 0.1001 while reducing the graph from 80 to 44 triples. DRKG similarly changes from 0.0410 to 0.0463 after soft support and retains 0.0463 after compression. On repoDB, fuzzy retrieval uses the original candidate order and raises adjusted E2E MRR from 0.0366 to 0.0378 with 55 rather than 100 triples. These comparisons show that a smaller evidence package can retain the adjusted ranking outcome; whether it also preserves paths, representations, or individual generated answers requires the separate analysis below.

These findings support two related but distinct conclusions. Soft support is most useful when the prepared graph contains query-specific features that distinguish candidates, as on PrimeKG and PharmKG and, more modestly, DRKG. Its effect is weaker when support is sparse or graph topology differs, as in repoDB and parts of Hetionet. Fuzzy retrieval primarily improves evidence efficiency. By design, it preserves the candidate MRR of its input row, while reducing graph size by approximately 45% in every fuzzy setting. The graph adapter consequently receives a smaller and more inspectable context, and retained evidence is easier to audit. The repoDB result indicates relevance to a clinical benchmark rather than prospective clinical validation: the dataset includes approved pairs and failed-like evidence, but its local UMLS disease nodes limit query-specific mechanistic support.

4.5 Automatic calibration and coefficient transfer

To test dependence on manually assigned dataset-specific weights, we evaluate a common five-feature candidate scorer with equal weights, learned weights, and leave-one-dataset-out (LODO) coefficients. The features describe original rank, graph contact, degree, shared neighbors, and direct query contact. Learned weights minimize a regularized pairwise logistic objective on validation data, with query-group cross-validation for regularization. LODO excludes all target-dataset calibration labels and selects coefficients using the other four datasets. Candidate membership and the upstream retriever remain fixed. Online Resource 1, Supplementary Section S12, provides the feature definitions, fitting objective, native-formula search, transfer protocol, and full diagnostics.

Table 4 Candidate calibration on the complete test splits. MRR@20 is computed over all rows, with zero reciprocal rank for absent gold entities. Shared equal, learned, and LODO use the same features; comparison with the original soft-support formula also changes features. LODO is coefficient transfer without target calibration labels.

Dataset	Gold@20	Original soft	Shared equal	Learned	LODO
PrimeKG	0.240	0.12533	0.12519	0.12533	0.12610
PharmKG	0.092	0.02816	0.03188	0.03188	0.03186
Hetionet	0.440	0.09544	0.10127	0.11528	0.11358
DRKG	0.200	0.04656	0.06143	0.06050	0.05861
repoDB	0.214	0.04811	0.05737	0.06490	0.06490

Table 4 shows that shared learned calibration improves the candidate-MRR point estimate over the original formula on four datasets and matches it on PrimeKG. LODO coefficients yield higher point estimates than the original formula on all five datasets. This indicates that useful candidate ordering can be obtained without fitting target-specific support coefficients. The feature-matched equal-weight control is essential: it already improves four datasets relative to the original formula and has the highest point estimate on DRKG. Learned-minus-equal MRR is 0.01402 on Hetionet and 0.00754 on repoDB, but the exploratory query-cluster 95% interval excludes zero only on repoDB, where it is [0.00040, 0.01518]. These results support transfer of the shared scoring representation rather than a universal advantage for learning its coefficients.

We also select edge-score coefficients automatically from 12 predefined vectors per dataset while keeping candidate order and graph size fixed. Selection uses validation exact-answer accuracy, then adjusted MRR, with a fixed tie rule; a source-only macro-validation variant excludes target labels. The resulting adjusted MRR remains close to the original fuzzy selector, without a consistent exact-answer advantage. Candidate reranking likewise has mixed generation effects. Detailed accuracy tables, coefficient values, cross-dataset transfers, label-efficiency curves, and sensitivity analyses appear in Supplementary Section S12 and Tables S16–S24. These retrospective experiments assess coefficient transfer within existing graph pipelines, rather than zero-shot completion on unseen diseases or knowledge graphs.

4.6 Information retention and alternative graph selection

We quantify compression relative to each retrieved source graph while fixing the candidate identities and order. Every compressed alternative selects the same number of unique triples as the original fuzzy graph for that query. The comparison includes random sampling, coverage-constrained random sampling, native-score top-k, strict coverage-first selection, degree, breadth-first selection, personalized PageRank, shortest paths, and relation-stratified selection. We separately measure candidate incidence, query–candidate reachability, labeled path retention, encoder representations, and generated answers. Full definitions and results are provided in Supplementary Section S13.

Table 5 Three-hop reachability at matched unique-edge budgets. Reachability is the per-row fraction of source-reachable candidates still reachable within three edges. N counts eligible rows; absent source connectivity is excluded. Random is mean ± standard deviation over five seeds. Fuzzy candidate-incidence retention relative to the source is 1.0000 on every dataset.

Dataset	N	Fuzzy	Random	Shortest paths	Shortest coverage
PrimeKG	488	0.7737	0.7760 ± 0.0038	0.9985	0.9993
PharmKG	500	0.9010	0.4423 ± 0.0064	1.0000	1.0000
Hetionet	90	0.4835	0.4848 ± 0.0428	1.0000	0.8514
DRKG	386	0.8921	0.5043 ± 0.0170	1.0000	0.8199
repoDB	297	0.9546	0.5465 ± 0.0241	1.0000	0.8971

Fuzzy retrieval retains all candidate incidence available in its source graphs. At the same unique-edge budgets, it preserves more three-hop reachability than uniform random selection on PharmKG, DRKG, and repoDB (Table 5); PrimeKG and Hetionet are similar to random. Shortest-path selection preserves nearly all source connectivity, but sacrifices incidence coverage on Hetionet, DRKG, and repoDB. These

Table 6 PrimeKG diagnostic cases of candidate promotion and evidence compression. Base and Soft denote gold-drug ranks before and after soft-support reranking; Removed is the number of triples discarded by fuzzy retrieval. These cases audit rank and evidence changes rather than provide treatment recommendations.

Bucket	Query / gold drug	Base	Soft	Removed	Interpretation
Improved	lymphosarcoma / Vincristine	16	2	20	Soft support promotes an evidence-connected antineoplastic candidate, and fuzzy retrieval preserves the improved order.
Improved	lymphoma / Vincristine	17	3	20	The gold drug is promoted from the tail under related lymphoma terminology.
Improved	rheumatoid arthritis / Methotrexate	19	8	23	The benchmark gold drug moves upward, showing why Gold@20 and MRR@20 should be reported together.
Sparse	seborrheic dermatitis / Cortisone acetate	1	1	26	Soft scoring avoids strict filtering when graph evidence is incomplete.
Cleaner	chronic otitis media / Norfloxacin	4	4	38	Candidate order is preserved while the graph package is substantially reduced.
Cleaner	cholera / Fusidic acid	1	1	33	Fuzzy retrieval removes redundant or shortcut-heavy evidence.
Cleaner	botulism / Ofloxacin	12	10	30	Candidate ordering is mostly preserved with a cleaner evidence graph.

results identify a trade-off between retaining some evidence for each candidate and retaining connected paths to the query, rather than establishing a single best selector.

Graph size alone also overstates unique-information reduction when retrieved triples repeat. On PrimeKG, the mean source contains 59.932 edge occurrences but 38.094 unique triples; fuzzy retrieval retains 32.340 unique triples. Thus, 21.838 of the 27.592 removed occurrences are duplicates, while 5.754 are unique triples. The other four prepared test datasets contain no source duplicates. Supplementary Section S13 distinguishes these counts and reports exact length-two/three paths, relation distributions, budget curves, and eligible sample counts.

The representation and generation comparisons further separate structural retention from downstream behavior. Fuzzy/source generated-answer agreement is 78.8%, 80.0%, 100%, 100%, and 80.8% on PrimeKG, PharmKG, Hetionet, DRKG, and repoDB, respectively. Even complete answer agreement can coexist with changed query representations and lost paths. Exact-answer comparisons do not identify a universally superior selector; the full results and counterexamples are reported in Supplementary Tables S28–S29. Fuzzy retrieval is therefore best characterized as coverage-oriented compression with measured structural and downstream trade-offs, not lossless preservation of biomedical mechanisms.

4.7 Case studies

The case studies examine whether the numerical changes in Table 3 correspond to evidence changes that a biomedical reader can inspect. A correct graph-augmented LLM answer may arise in several ways: the candidate generator may already rank the gold drug highly, the LLM may select an already high-ranked candidate, or evidence calibration may promote a recoverable but low-ranked candidate. We therefore examine representative PrimeKG examples in terms of rank change, graph-size reduction, retained internal evidence, and external biomedical plausibility.

Table 6 groups the cases into three diagnostic categories. Improved cases test whether soft support promotes recoverable, low-ranked gold drugs to more visible positions. Sparse cases test whether graded scoring avoids brittle hard filtering when graph support is incomplete. Cleaner cases assess fuzzy graph reduction when candidate ranks change little. This taxonomy follows the division of labor within SoftFuse-KGC: soft support targets candidate order, whereas fuzzy retrieval targets evidence efficiency. Fig. 4 visualizes the two principal examples, and Table 7 reports external evidence identifiers for model-selected candidates.

The first principal case is lymphosarcoma / vincristine. Vincristine occurs in the raw DrKGC-style top-20 list but only at rank 16. Soft support promotes it to rank 2, after which fuzzy retrieval preserves the improved order while reducing the evidence subgraph from 45 to 25 triples. This example is informative because “lymphosarcoma” is a historical lymphoma-related term. Fig. 4(a) shows that fuzzy retrieval

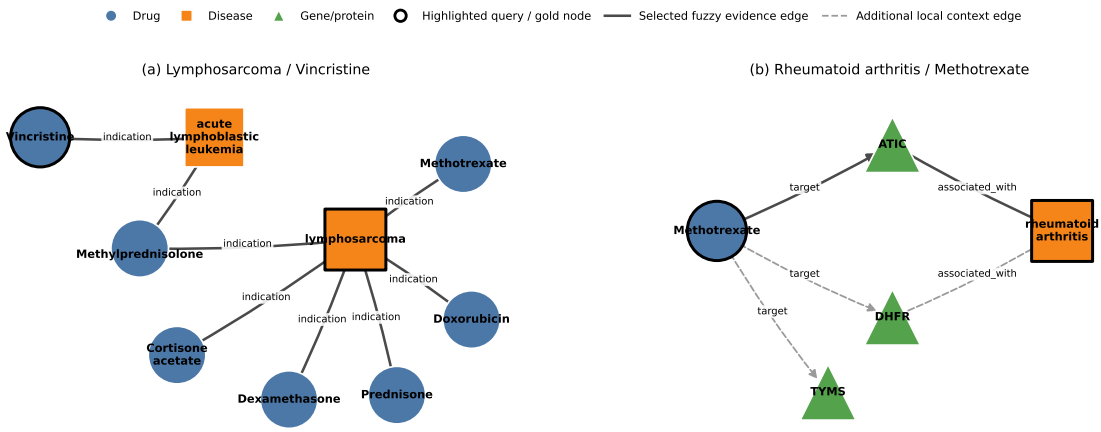


Fig. 4 Case-study evidence graphs for lymphosarcoma / Vincristine and rheumatoid arthritis / Methotrexate. Circles denote drugs, squares denote diseases, and triangles denote genes or proteins. Thick solid edges show evidence selected by fuzzy retrieval, whereas dashed edges show additional local context. Black outlines identify the query disease and gold drug, making rank changes and retained evidence auditable

Table 7 Internal graph evidence and external plausibility evidence for case-study candidates. Candidate drugs come from the corresponding SoftFuse-KGC ranked lists, and internal evidence is extracted from the fuzzy-retrieved graph. The external evidence column provides a literature citation and PubMed identifier for candidate–disease plausibility or related clinical context. These references support plausibility checking and auditability, not prospective clinical validation, graph-edge provenance, or treatment recommendations.

Case disease	Candidate drug	Model rank / band	Internal SoftFuse-KGC evidence	External plausibility evidence
Lymphosarcoma	Prednisolone	Rank 1 / top	Prednisolone – indication – mycosis fungoides and variants	[57]; PMID:8422281
Lymphosarcoma	Vincristine	Rank 2 / top	Vincristine – indication – acute lymphoblastic leukemia	[58]; PMID:5066691
Lymphosarcoma	Dexamethasone	Rank 5 / top	Dexamethasone – indication – lymphosarcoma; reverse direct edge retained	[59]; PMID:8120556
Lymphosarcoma	Doxorubicin	Rank 12 / tail	Doxorubicin – indication – lymphosarcoma	[60]; PMID:7680764
Lymphosarcoma	Bleomycin	Rank 15 / tail	Bleomycin – indication – lymphosarcoma	[61]; PMID:1384144
Rheumatoid arthritis	Methotrexate	Rank 8 / mid	Methotrexate – target – ATIC; ATIC – associated_with – rheumatoid arthritis	[62]; PMID:34101387
Rheumatoid arthritis	Dexamethasone	Rank 11 / tail	Dexamethasone – indication – rheumatoid arthritis	[63]; PMID:4158821
Rheumatoid arthritis	Hydrocortisone	Rank 13 / tail	Hydrocortisone – indication – rheumatoid arthritis	[64]; PMID:8173851
Rheumatoid arthritis	Prednisolone	Rank 16 / tail	Prednisolone – indication – rheumatoid arthritis	[65]; PMID:35641125
Rheumatoid arthritis	Prednisone	Rank 18 / tail	Prednisone – indication – rheumatoid arthritis	[66]; PMID:18207016

retains the promoted vincristine context and additional lymphoma-relevant local evidence while removing redundant triples.

The second principal case is rheumatoid arthritis / methotrexate. Clinical guidance, labeling, and clinical studies identify methotrexate as an established rheumatoid arthritis therapy [62, 67, 68]. The raw candidate generator retrieves methotrexate at rank 19, a weak position for candidate-constrained generation and manual review. Soft support raises it to rank 8, and fuzzy retrieval preserves that rank while reducing the graph from 52 to 29 triples. Fig. 4(b) retains the mechanistic Methotrexate–ATIC–rheumatoid arthritis path together with additional target and association context. The example illustrates why Gold@20 and MRR@20 should be considered together: the raw list contains the gold drug, but its position limits practical utility.

The external evidence in Table 7 provides a plausibility check for model-selected candidates rather than provenance for every internal graph edge. In the lymphosarcoma case, the table includes the promoted gold drug Vincristine and other lymphoma-context candidates ranked by the model. In the rheumatoid arthritis case, methotrexate is the gold candidate, whereas the listed steroids represent adjunctive anti-inflammatory context. The candidates are not claimed to be therapeutically equivalent.

The remaining cases illustrate complementary behavior. Cortisone acetate remains at rank 1 for seborrheic dermatitis and shows why binary filtering can be risky when graph support is incomplete. The chronic otitis media / norfloxacin, cholera / fusidic acid, and botulism / ofloxacin examples show that fuzzy retrieval can remove redundant or shortcut-heavy triples while preserving candidate order. Because graph proximity does not itself establish therapeutic appropriateness, all cases require cautious biomedical interpretation. Their purpose is to show that SoftFuse-KGC makes rank and evidence changes auditable, not to provide clinical validation.

5 Discussion

The results support soft, confidence-aware evidence modeling for biomedical graph-augmented LLM completion. More specifically, they show that candidate ordering and graph-evidence selection are separate problems. Soft support can improve ranking when the available evidence differentiates candidates, whereas fuzzy retrieval can reduce graph size without changing the candidate list. This distinction matters in biomedical informatics because ranking performance, evidence efficiency, and interpretability all affect the usefulness of a graph-LLM pipeline.

Evaluation across five datasets shows that the benefit of soft support depends on the evidence available for each query. The shared-feature calibration and leave-one-dataset-out analyses demonstrate that reranking need not rely exclusively on separately chosen manual coefficients. Equal weights already improve several candidate-ranking point estimates, however, and learned coefficients provide no uniform generation benefit. For graph compression, preserving candidate incidence and adjusted MRR does not imply preservation of all multi-hop evidence. Shortest-path selection retains connectivity more effectively, whereas fuzzy retrieval retains broader candidate incidence on several datasets. The choice of selector should therefore reflect the intended evidence-review objective.

The structure-only comparisons further define the scope of the contribution. R-GCN serves as the lightweight source because it matches the DrKGC graph-adapter interface and provides relation-aware local embeddings for the subgraph encoder. It is not assumed to be the strongest standalone candidate generator. The calibrated graph-compatible row is strongest or among the strongest in candidate MRR on PrimeKG and PharmKG, whereas HRGAT or DistMult performs better on Hetionet, DRKG, and repoDB. These results indicate the limitation imposed by the selected source but also illustrate the modularity of SoftFuse-KGC. A future implementation could replace R-GCN with a stronger structural or hybrid retriever that exposes compatible entity embeddings, candidate lists, and query-specific evidence graphs.

repoDB is a clinically relevant external benchmark rather than clinical validation. Its approved drug–indication pairs and failed-like diagnostic evidence are useful for evaluation, but its local UMLS disease nodes do not map directly to the reused DRKG disease namespace. The shared calibration features improve candidate ranking in this sparse alignment setting, and fuzzy retrieval reduces the average evidence package from 100 to 55 triples. The alternative-selector comparison nevertheless contains better exact-generation point estimates than fuzzy retrieval (Supplementary Section S13). This supports a bounded claim about evidence organization rather than optimal downstream prediction.

The case studies make these behaviors more interpretable, but they require careful biomedical framing. The lymphosarcoma / vincristine example combines a large rank improvement with retained lymphoma-related evidence. The rheumatoid arthritis / methotrexate example combines a rank improvement with guideline, labeling, clinical-study, and graph-path context. These examples demonstrate candidate

reranking and evidence packaging rather than clinical validity. SoftFuse-KGC should therefore be used for prioritization, hypothesis generation, and evidence review. It does not replace clinical judgment, pharmacological assessment, or experimental validation.

The current study has several limitations. First, the fixed top-20 candidate set bounds recoverability. Second, the new calibration study transfers coefficients within prepared benchmark pipelines; it does not establish zero-shot completion on unseen graphs or diseases. Third, edge budgets and the original feature families remain design choices even when coefficients are selected automatically. Fourth, higher candidate MRR and retained structural connectivity do not consistently yield more exact generated answers. Fifth, target-relation semantics and entity alignment differ across datasets, and the retained paths are structural witnesses rather than expert-validated therapeutic mechanisms. The analyses are retrospective, and bootstrap intervals are exploratory rather than multiplicity-adjusted confirmatory tests.

The spatial–spectral and frequency-domain multi-source models discussed in Section 2 suggest a route beyond fixed evidence heuristics [45, 46]. Molecular, interaction, and pathway representations could supply complementary features to a learned candidate scorer or budget-constrained edge selector. A future extension could jointly optimize predictive utility and retention of source-supported paths, with source-wise gating to handle missing or conflicting modalities. Such integration would require consistent entity alignment and held-out evaluation of both candidate coverage and generated answers; the present topology-based calibration is a limited first test, not an implementation of those multi-source architectures.

6 Conclusion

We presented SoftFuse-KGC as an inspectable combination of candidate reranking and graph-evidence selection for biomedical graph-augmented LLM completion. Across five datasets, the experiments separate candidate recoverability, ranking, structural retention, and generated-answer behavior. Shared learned and transferred coefficients provide alternatives to dataset-specific support weights, while equal-weight controls and downstream evaluation show where their advantages are limited. Matched-budget graph comparisons characterize fuzzy retrieval as coverage-oriented compression with topology-dependent path and representation losses. The contribution is this biomedical interface design and its measured trade-offs, rather than a general claim of new reranking or lossless compression principles.

Future work will pursue several directions. First, we will integrate stronger structural and hybrid candidate retrievers to improve candidate coverage while retaining compatibility with the graph-adapter interface. Second, we will extend the evaluated linear calibration to uncertainty-aware multi-source fusion and query-adaptive graph budgets. Third, we will strengthen cross-dataset entity and ontology alignment, particularly between repoDB UMLS concepts and mechanistic graph namespaces, and extend the evaluation to additional therapeutic relations. Fourth, learned evidence retrievers will be studied to balance graph compression against end-to-end performance across different graph topologies. Finally, temporal evaluation, broader language-model comparisons, and domain-expert assessment will be used to examine whether the retained evidence remains robust, mechanistically informative, and clinically plausible.

Supplementary Information. Additional details are provided in the Supplementary Material. The opening *The Web Application* section describes the public demo, screenshots, and tab-level usage notes. We provide a public web application for interactive inspection of the SoftFuse-KGC pipeline at https://inosuke710-softfuse-kgc-demo.hf.space/. It uses precomputed artifacts and allows users to select a dataset, split, disease or query case, and top-k value before viewing the Overview, Soft Support, and Fuzzy Retrieval tabs. Across tabs, the interface links a selected case to summary metrics, candidate-level support features, rank changes, and retained evidence edges. Users can therefore trace a reranking or graph-compression result to the final evidence package. Sections S1–S3 report feature selection, weight assignment, frozen soft-support coefficients, fuzzy edge-scoring weights, and graph-budget checks. Sections S4–S6 provide the inference algorithm, graph-LLM settings, prompt and retrieval-rule templates, and

PrimeKG formula variants. Sections S7–S9 report LLM comparisons, decoding checks, ablation, sensitivity, robustness, and the fuzzy encoder probe. Sections S10–S11 provide structure-only baseline training settings, dataset preprocessing, and leakage checks. Tables S1–S15 contain the corresponding configuration and diagnostic summaries. Section S12 adds automatic calibration, learned coefficients, cross-dataset transfer, label-efficiency, sensitivity, and generated-answer comparisons (Tables S16–S24). Section S13 adds matched-budget selectors, structural-retention metrics, budget curves, representation changes, and generated-answer comparisons (Tables S25–S29 and Figs. S4–S6).

Statements and Declarations

Competing interests

The authors have no relevant financial or non-financial interests to disclose.

Funding

Not applicable.

Ethics statement

This study used publicly available biomedical knowledge graph resources and involved no human participants, human tissue, animal experiments, or identifiable patient-level data.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used ChatGPT (OpenAI) for assistance with grammar checking, language polishing, and minor improvements to sentence structure. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article.

Author contributions

Le Van Thanh: Conceptualization, Methodology, Investigation, Resources, Writing-original draft, Validation, Writing-review and editing; Dang Hoang Duong: Conceptualization, Methodology, Investigation, Writing-original draft, Visualization, Software, Hai Van Pham: Writing-review and editing, Validation, Supervision, Project administration.. All authors read and approved the final manuscript.

Data availability

The datasets are available from their original public sources.

PrimeKG: https://doi.org/10.7910/DVN/IXA7BM (raw CSV: https://dataverse.harvard.edu/api/access/datafile/6180620);

PharmKG: https://github.com/biomed-AI/PharmKG and https://zenodo.org/records/4077338;

Hetionet v1.0: https://zenodo.org/records/268568;

DRKG at https://github.com/gnn4dr/DRKG (archive: https://dgl-data.s3-us-west-2.amazonaws.com/dataset/DRKG/drkg.tar.gz);

repoDB: https://github.com/adam-sam-brown/repoDB.

Code availability

The source code, figure-generation scripts, processed artifacts, and reproduction instructions are available at https://github.com/dhduong710/Project_1.

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